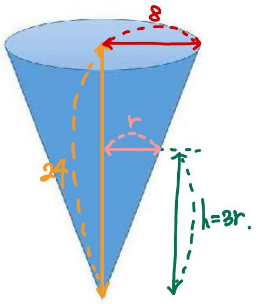
1



$v = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \cdot \left(\frac{h^2}{9}\right) \cdot h = \frac{\pi}{27}h^3.$

$\underbrace{\frac{dV}{dt}}_{=3t^2} = \underbrace{\frac{\pi}{9}h^2}_{=\frac{\pi}{9}\cdot(2\pi)^2 = \frac{4}{9}\pi^2} \cdot \frac{dh}{dt}.$ $\Rightarrow \frac{dh}{dt} \overset{t=a}{=} 3a^2 \cdot \frac{9}{4\pi^2}$

$v = \int_0^a 3t^2 dt = a^3 = \frac{\pi}{27}\cdot 8\pi^3 = \frac{8}{27}\pi^4$ $\quad \therefore t = a = \frac{2}{3}\pi^{\frac{4}{3}}.$

$\therefore \frac{dh}{dt} = 3a^2 \cdot \frac{9}{4\pi^3} = \frac{3}{\sqrt[3]{\pi}}$ //

2

$x^2+(y-\frac{5}{4})^2=r^2$ & $y=x^2$ 과 내접.

연립 $\rightarrow$ $y+(y-\frac{5}{4})^2=r^2 \iff y^2-\frac{3}{2}y+\left(\frac{25}{16}-r^2\right)=0$

이때. $D=0$을 만족!

$D=\frac{9}{4}-4\left(\frac{25}{16}-r^2\right)=0 \iff r=1 \ (r>0).$

3

$\left|\frac{(x+1)\sin(x^2)}{x^2+1}\right| \le \frac{x+1}{x^2+1}$

$x\to\infty \Downarrow$ $\quad \downarrow x\to\infty$

0 (∵ 조임정리) $\quad$ 0

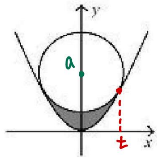
4

$\Rightarrow \lim_{\alpha\to\infty}\left[\underbrace{\ln\left|x+\sqrt{x^2+9}\right| - a\ln|x+1|}_{=\ln\left|\frac{x+\sqrt{x^2+9}}{(x+1)^a}\right|}\right]_0^{\alpha} \xrightarrow{a=1} I=\ln 2-\ln 3=\ln\frac{2}{3}$

$a=1.\ b=1.\ c=2.\ d=3$ $\quad \therefore a+b+c-d=1$ //

5

원: $x^2+(y-a)^2=1$ & $y=x^2$과 연립. (원 하단부: $y=a-\sqrt{1-x^2}$)



$\Rightarrow$ 미분계수 Same : $\frac{t}{\sqrt{1-t^2}}=2t \quad (t\neq 0).$

$\therefore t=\frac{\sqrt{3}}{2}.$

$\cdot\ (0,a) \sim \left(\frac{\sqrt{3}}{2},\frac{3}{4}\right)$ 의 거리 $=\frac{3}{4}+\left(a-\frac{3}{4}\right)^2=1 \quad \therefore a=\frac{5}{4}$

$\therefore 2\cdot\int_0^{\frac{\sqrt{3}}{2}}\left(\frac{5}{4}-\sqrt{1-x^2}-x^2\right)dx = \frac{3}{4}\sqrt{3}-\frac{\pi}{3}$ //

6

$\int \frac{\boxed{\sin^7 x}}{\cos^4 x}dx$ $= \sin x\cdot\sin^6 x$ $\quad = \int \frac{\sin x}{\cos^4 x}(1-\cos^2 x)^3 dx$

$\cos x \overset{\triangle}{=} t \Rightarrow -\sin x\,dx=dt.$

$\int -\frac{1}{t^4}\underbrace{(1-t^2)^3}_{=1-3t^2+3t^4-t^6}dt = \int\left(t^2-3+\frac{2}{t^2}-\frac{1}{t^4}\right)dt$

$=\frac{1}{3}t^3-3t-2t^{-1}+\frac{1}{3}t^{-3}+C = \frac{1}{3}\cos^3 x-3\cos x-\frac{2}{\cos x}+\frac{1}{3\cos^3 x}+C$ //

7

$L_f = \underbrace{f(0)}_{=2-ab}+\underbrace{f'(0)}_{=a-b.}(x-0). = (a-b)x+(2-ab)$
$= 2x-1$ $\quad \therefore a=3.\ b=1.$

$f'(x)=ae^{ax}-be^{-bx}\Big|_{x=0}=a-b$

$\therefore f''(x)=a^2e^{ax}+b^2e^{-bx}\Big|_{x=0}=a^2+b^2=10$ //

8

$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\frac{(x-a)^{n+2}}{3^{2n+4}}}{\frac{(x-a)^{n+1}}{3^{2n+2}}}\right| = \left|\frac{x-a}{9}\right| (<1) \quad \therefore |x-a|<9.$

$\Rightarrow -9+a<x<a+9 \Rightarrow a=5$

$\therefore f(1)=\sum_{n=0}^{\infty}\frac{(-4)^{n+1}}{3^{2n+2}} = -\frac{4}{9}\cdot\frac{1}{1-\left(-\frac{4}{9}\right)} = -\frac{4}{13}$ //

9

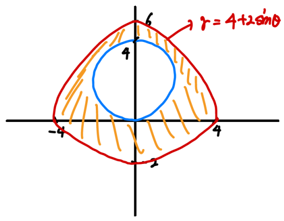
$x-1 \overset{\triangle}{=} t$

$f(t) = (t+1)^3 \cdot e^{t+1} = (t^3+t^2+t+1)e^{t+1}$.

$= e \cdot (t^3+t^2+t+1) \cdot \underbrace{\sum_{n=0}^{\infty} \frac{t^n}{n!}}_{\frac{t^{n-3}}{(n-3)!} + \frac{t^{n-2}}{(n-2)!} + \frac{t^{n-1}}{(n-1)!} + \frac{t^n}{n!}}$

$\therefore f^{(n)}(0) = e \cdot \underbrace{\left(\frac{1}{(n-3)!} + \frac{1}{(n-2)!} + \frac{1}{(n-1)!} + \frac{1}{n!} \right) \cdot n!}_{n(n-1)(n-2) + n(n-1) + n + 1} = e(n^3+2n+1)$ //

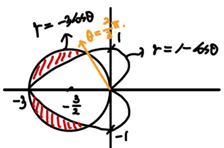
10



$A = \frac{1}{2}\int_0^{2\pi} \underbrace{(4+2\sin\theta)^2}_{16+16\sin\theta+4\sin^2\theta} d\theta - 4\pi$

$= \frac{1}{2} \cdot \left(32\pi + 4 \cdot 4 \cdot \frac{1}{2} \cdot \frac{\pi}{2}\right) - 4\pi = 16\pi + 2\pi - 4\pi = 14\pi$ //

11



$2 \times \frac{1}{2} \int_{\frac{2}{3}\pi}^{\pi} \{ \underbrace{(9\cos^2\theta) - (1-\cos\theta)^2}_{9\cos^2\theta - 1 + 2\cos\theta - \cos^2\theta = 8\cos^2\theta + 2\cos\theta - 1} \} d\theta$, $\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$= \left[4\theta + 2\sin 2\theta + 2\sin\theta - \theta \right]_{\frac{2}{3}\pi}^{\pi} = (4\pi - \pi) - \left(\frac{8}{3}\pi - \sqrt{3} + \sqrt{3} - \frac{2}{3}\pi\right) = \pi$.

$\therefore a = 1 . b = 0 \Rightarrow 11a^2 + b^2 = 11$ //

12

$\begin{vmatrix} \nabla f \\ \nabla g \\ \nabla h \end{vmatrix} = 0$ 이용!

$\begin{vmatrix} 2x & 2y & 2z \\ 1 & 1 & -1 \\ -2x & -2y & 2z \end{vmatrix} = \begin{vmatrix} 0 & 2(y-x) & 2(x+z) \\ 1 & 1 & -1 \\ 0 & 2(x-y) & 2(z-x) \end{vmatrix}$

$= (-1) \cdot \left(4(y-x)(z-x) + 4(y-x)(x+z) \right)$

$= -4(y-x) \cdot 2z = 0$

$\therefore z = 0$ or $x = y$.

i) $x = y$.

$2x^2 = z^2$ & $2x - z + 1 = 0 \Rightarrow f = 2x^2 + z^2 = 4x^2$.

$z = 2x+1$

$\hookrightarrow 2x^2 = 4x^2 + 4x + 1 \quad \therefore 2x^2 + 4x + 1 = 0$.

$\Rightarrow x = \frac{-2 \pm \sqrt{2}}{2} \quad \therefore x^2 = \begin{cases} \frac{6 - 4\sqrt{2}}{4} \\ \frac{6 + 4\sqrt{2}}{4} \end{cases} \Rightarrow 4x^2 = \begin{cases} 6 - 4\sqrt{2} \\ 6 + 4\sqrt{2} \end{cases}$.

ii) $z = 0$.

$x^2 + y^2 = 0$, $x + y - 1 = 0 \Rightarrow f = x^2 + y^2 = 0$ //

$\therefore M + m = 12$ //

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$f_x(0.0) = \lim_{h \to 0} \frac{f(0+h, 0) - f(0.0)}{h} = 0$.

$f_x = \frac{y^2(x^2+y) - xy^2 \cdot 2x}{(x^2+y)^2}$

$f_{xy}(0,0) = \lim_{h \to 0} \frac{f_x(0, 0+h) - f_x(0.0)}{h} = 1$

14

$f = g(\underbrace{y\cos x + e^x}_{u}, \underbrace{4y^2}_{v}) \Rightarrow f = g\langle \begin{smallmatrix} u \\ v \end{smallmatrix} \rangle x . y$

$f_y = g_u \cdot \underbrace{u_y}_{\cos x} + g_v \cdot \underbrace{v_y}_{8y} \Big|_{(x,y) = (0,-1)} = (2) \cdot 1 + (-8) \cdot 1 = -6$ //

$(x,y) = (0,-1) \Rightarrow (u,v) = (0,4)$

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오두 동일!

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$xy \triangleq u \,.\; \frac{y}{x} \triangleq v \quad \Rightarrow \left| J(u,v) \right| = \frac{1}{2v}$

$\therefore \int_{\frac{1}{4}}^{2} \int_{1}^{4} \frac{1}{2v} \cdot e^{-\frac{u}{2}} \, du\, dv = \left(e^{-\frac{1}{2}} - e^{-2}\right) \ln 8$

17

$A = \int_0^{2\pi} \int_0^{2} \int_r^{4-r} r^2 \cdot dz\,dr\,d\theta = \frac{16}{3}\pi$

$B = \int_0^{2\pi} \int_0^{1} \int_r^{2-r} r^2 \cdot dz\,dr\,d\theta = \frac{\pi}{3}$

$\ominus \Rightarrow 5\pi$

18

$r_u = (1, -1, 1)$

$r_v = (0, 2v, 1)$

$\Rightarrow r_u \times r_v = \begin{pmatrix} -1 & 1 & 1 & -1 \\ 2v & 1 & 0 & 2v \end{pmatrix} = (-1-2v, -1, 2v).$

$\Rightarrow \int_0^4 \int_0^3 \langle -v^2+u, u, 0 \rangle \cdot \langle -1-2v, -1, 2v \rangle \, du\, dv = 340$

19

$N_x - M_y = \left(\frac{3}{x} y^2 - 2xz\right) - \left(\frac{3y^2}{x} - 2xz\right) = 0$

$f = y^3 \ln x - x^2 yz$ (potential 함수).

선·기본정리 $\left[y^3 \ln x - x^2 yz \right]_{(1, e, \frac{1}{e})}^{(e, 1, 0)} = (1) - (-1) = 2$

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$r = \langle 2\cos\theta, 2\sin\theta \rangle \left(\frac{\pi}{6} \le \theta \le \frac{5}{6}\pi\right).$

$\therefore \int_{\frac{\pi}{6}}^{\frac{5}{6}\pi} \left\langle \frac{2\sin\theta}{5}, \frac{2\cos\theta}{5} \right\rangle \cdot \langle -2\sin\theta, 2\cos\theta \rangle \, d\theta$

$\frac{4}{5}\left(\cos^2\theta - \sin^2\theta\right) = \frac{4}{5}\cos 2\theta$

$= \frac{4}{5} \left[\frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{6}}^{\frac{5}{6}\pi} = \frac{2}{5}\left(-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}\right) = -\frac{2}{5}\sqrt{3}$